CSCE 763: Digital Image Processing
Homework #4

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Problem 1: Highboost Filtering (10 pts)

As we discussed in class, image sharpening by unsharp masking consists of three steps. Design a 3×3 filter for performing highboost filtering with $k = 3$ in a single pass through an image, a single 3×3 highboost filter performing image sharpening. Assume that the average image is obtained using the filter shown below:

$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Step 1: Highboost Formula

Let f be the input image and $\bar{f} = f \otimes h_{\text{avg}}$ be the blurred image. The highboost form is

$$g = f + k(f - \bar{f}) = (1 + k)f - k\bar{f}$$

For $k = 3$:

$$g = 4f - 3(f \otimes h_{\text{avg}}) = f \otimes (4\delta - 3h_{\text{avg}})$$

So the required single-pass kernel is

$$h_{\text{HB}} = 4\delta - 3h_{\text{avg}}$$

Step 2: Compute the 3×3 Coefficients

Given

$$h_{\text{avg}} = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad \Rightarrow \quad 3h_{\text{avg}} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

The impulse term 4δ contributes 4 only at the center, 0 elsewhere. Therefore:

$$h_{\text{HB}} = \begin{bmatrix} -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & 4 - \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \end{bmatrix} = \begin{bmatrix} -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{11}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \end{bmatrix}$$

Equivalent scaled form:

$$h_{\text{HB}} = \frac{1}{3}\begin{bmatrix} -1 & -1 & -1 \\ -1 & 11 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Results

The required single-pass 3×3 highboost filter for $k = 3$ is

$$\boxed{h_{\mathrm{HB}} = \begin{bmatrix} -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{11}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} -1 & -1 & -1 \\ -1 & 11 & -1 \\ -1 & -1 & -1 \end{bmatrix}}$$

Problem 2: Fourier Transform (20 pts)

Perform Fourier Transform on the function

$$f(t) = \begin{cases} 14, & -2w \le t \le 0 \\ 3, & 0 \le t \le 2w \\ 0, & \text{otherwise} \end{cases}$$

Step 1: Use the Continuous FT Definition

Using the convention

$$F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut}\,dt$$

we get

$$F(u) = 14\int_{-2w}^{0} e^{-j2\pi ut}\,dt + 3\int_{0}^{2w} e^{-j2\pi ut}\,dt$$

Step 2: Evaluate the Two Integrals

For $u \neq 0$:

$$14\int_{-2w}^{0} e^{-j2\pi ut}\,dt = \frac{14\,(e^{j4\pi uw} - 1)}{j2\pi u}$$

$$3\int_{0}^{2w} e^{-j2\pi ut}\,dt = \frac{3\,(1 - e^{-j4\pi uw})}{j2\pi u}$$

Hence

$$F(u) = \frac{14e^{j4\pi uw} - 3e^{-j4\pi uw} - 11}{j2\pi u}$$

Step 3: Equivalent Trigonometric Form

Let $\alpha = 4\pi uw$. Then

$$14e^{j\alpha} - 3e^{-j\alpha} - 11 = 11(\cos\alpha - 1) + j17\sin\alpha$$

So

$$F(u) = \frac{17\sin(4\pi uw) + j11\,[1 - \cos(4\pi uw)]}{2\pi u}, \quad u \neq 0$$

At $u = 0$, from direct area under $f(t)$:

$$F(0) = \int f(t)dt = 14(2w) + 3(2w) = 34w$$

Results

$$\boxed{F(u) = \frac{14e^{j4\pi uw} - 3e^{-j4\pi uw} - 11}{j2\pi u},\ u \neq 0; \qquad F(0) = 34w}$$

Equivalent form:

$$\boxed{F(u) = \frac{17\sin(4\pi uw) + j11\left[1 - \cos(4\pi uw)\right]}{2\pi u},\ u \neq 0}$$

Problem 3: Mean/Order-Statistic Filters (30 pts)

The white bars in the test pattern shown in Figure 1 are 7 pixels wide and 210 pixels high. The separation between bars is 17 pixels. What would this image look like after application of a 9×9 filter of:

(a) Arithmetic mean filter

(b) Geometric mean filter

(c) Harmonic mean filter

(d) Contraharmonic mean filter with $Q = 1$

(e) Contraharmonic mean filter with $Q = 0$

(f) Contraharmonic mean filter with $Q = -1$

(g) Median filter

(h) Max filter

(i) Min filter

(j) Midpoint filter

Hints: you can assume the black background has intensity 0 and white bars have intensity 255. You can either write a program to implement these filters or represent the image as an array with numbers and show the results after filtering. You may need to pay attention to the corners of the white bars.

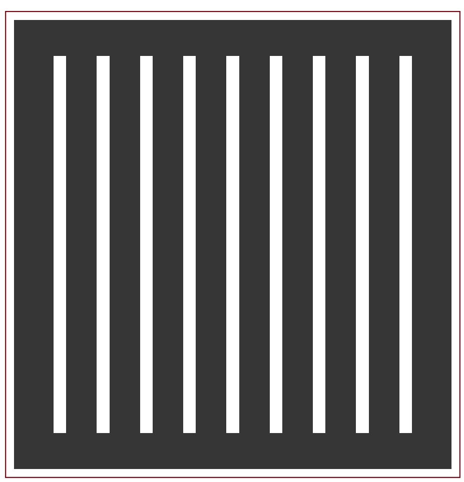

Figure 1: Figure 1

Step 1: 1D Scan-Line Model

Analyze one horizontal scan line through bar interiors. Intensities are binary: 255 inside bars and 0 in background. Bar width = 7, gap = 17, period $T = 24$. The 9×9 mask has half-width 4 in each direction. Because $7 < 9$, no window can be entirely white in the horizontal direction.

Step 2: Useful Counts for Binary Windows

For a binary neighborhood containing n_w white pixels and $81 - n_w$ black pixels:

$$\text{arithmetic mean} = \frac{255 n_w}{81}$$

$$\max = \begin{cases} 255, & n_w > 0 \\ 0, & n_w = 0 \end{cases}, \qquad \min = \begin{cases} 255, & n_w = 81 \\ 0, & n_w < 81 \end{cases}$$

$$\text{midpoint} = \frac{\max + \min}{2}$$

For bar-interior rows, maximum possible n_w is $7 \times 9 = 63$.

Filter-by-Filter Outcomes

(a) Arithmetic mean: Peak value at bar center is $\frac{63}{81} \cdot 255 \approx 198.3$. Bars become blurred gray ridges; gaps remain dark in the center (since $17 - 8 = 9$ columns in each gap never touched by the window).
(b) Geometric mean: Any zero in window makes the product zero. Every 9×9 window contains at least one zero because bar width is only 7. Output is all black.
(c) Harmonic mean: Any zero in the window makes denominator infinite, so output is 0. Output is all black.
(d) Contraharmonic mean, $Q = 1$:

$$\hat{f} = \frac{\sum z^2}{\sum z}$$

For binary values, if $n_w > 0$, this equals 255; if $n_w = 0$, equals 0. This is equivalent to a max filter: bars are dilated by 4 pixels on each side (width $7 \to 15$), still separated because gap becomes 9.
(e) Contraharmonic mean, $Q = 0$: Reduces to arithmetic mean; same result as (a).
(f) Contraharmonic mean, $Q = -1$: Requires division by $\sum z^{-1}$; zeros cause singular behavior and drive output to 0 whenever any black pixel exists. Since every window contains black pixels, output is effectively all black.
(g) Median filter: Window size is 81, so median is white iff at least 41 values are white. On interior bar rows, this occurs only where horizontal overlap with the bar is at least 5 columns, which preserves an approximately 7-pixel bar width. Bars remain separated and become cleaner at edges.
(h) Max filter: Same as (d): binary dilation, width $7 \to 15$, gap $17 \to 9$.
(i) Min filter: Needs all 81 values white. Impossible here because bar width < 9. Output all black.
(j) Midpoint filter: Since all-white windows never occur, min is always 0. Where any white is present, max is 255, so midpoint is 127.5. Result is gray dilated bars (value about 128) on black background.

Computed 9x9 Filter Outputs

The exact filter outputs computed from the binary 0/255 test pattern are shown below.

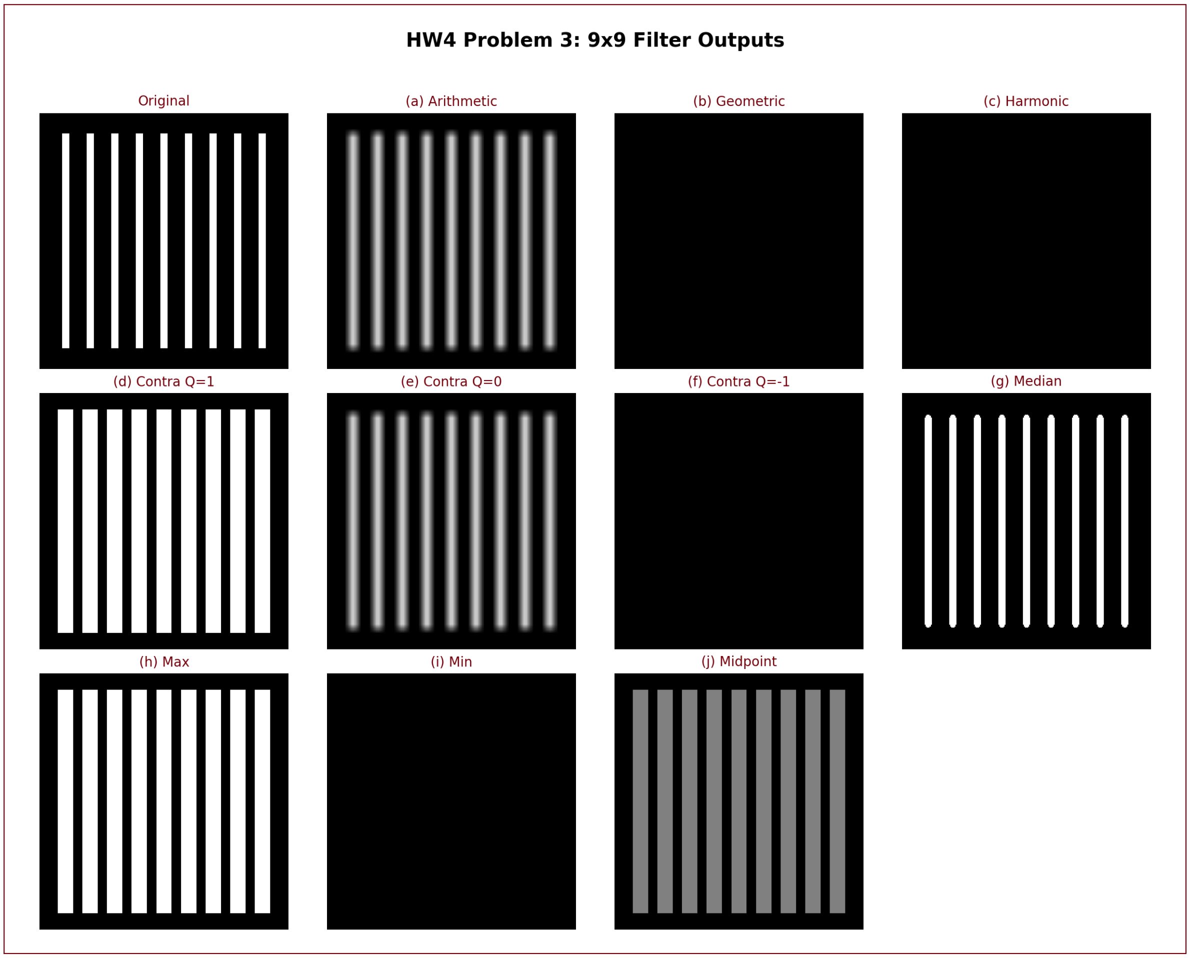

Figure 2: Problem 3 computed outputs for filters (a)–(j)

Results

Filter	Result on This Pattern
Arithmetic mean	Blurred gray bars, peak ≈ 198, bars remain separated.
Geometric mean	All black (0 everywhere).
Harmonic mean	All black (0 everywhere).
Contraharmonic ($Q = 1$)	Binary dilation (same as max), width $7 \to 15$, gap $17 \to 9$.
Contraharmonic ($Q = 0$)	Same as arithmetic mean.
Contraharmonic ($Q = -1$)	Effectively all black due to zeros in every window.
Median	Approx. original bar width preserved; noise-smoothed edges, separation preserved.
Max	Same as $Q = 1$: binary dilation.
Min	All black (requires all-255 window, impossible).
Midpoint	Gray dilated bars (≈ 127.5 where window sees white), black elsewhere.

Problem 4: Motion Blur Model (20 pts)

During acquisition, an image undergoes uniform linear motion in the horizontal direction (y-direction) for a time interval T_1. The direction of motion then switches to the vertical direction (x-direction) for a time interval T_2. Assuming that the time it takes the image to change directions is negligible, and that shutter opening time and closing time are also negligible, give an expression for the blurring function, $H(u, v)$.
(Hint: you can refer to the example that the blurring function for motion during acquisition is

$$H(u, v) = \int_0^T e^{-j2\pi[u\,x_0(t)+v\,y_0(t)]}\,dt$$

)

Step 1: Piecewise Motion Model

Let the first segment move along y with constant speed b for $0 \le t \le T_1$, and the second segment move along x with constant speed a for $T_1 < t \le T_1 + T_2$:

$$x_0(t) = 0,\ y_0(t) = bt, \quad 0 \le t \le T_1$$

$$x_0(t) = a(t - T_1),\ y_0(t) = bT_1, \quad T_1 < t \le T_1 + T_2$$

Step 2: Split the Integral Over the Two Time Intervals

$$H(u, v) = \int_0^{T_1} e^{-j2\pi vbt}\,dt + \int_{T_1}^{T_1+T_2} e^{-j2\pi[ua(t-T_1)+vbT_1]}\,dt$$

Step 3: Closed Form Expression

Evaluating each term:

$$\int_0^{T_1} e^{-j2\pi vbt}\,dt = \frac{1 - e^{-j2\pi vbT_1}}{j2\pi vb}$$

$$\int_{T_1}^{T_1+T_2} e^{-j2\pi[ua(t-T_1)+vbT_1]}\,dt = e^{-j2\pi vbT_1}\,\frac{1 - e^{-j2\pi uaT_2}}{j2\pi ua}$$

Therefore

$$H(u, v) = \frac{1 - e^{-j2\pi vbT_1}}{j2\pi vb} + e^{-j2\pi vbT_1}\frac{1 - e^{-j2\pi uaT_2}}{j2\pi ua}$$

(with limiting values for $u = 0$ or $v = 0$).

Results

A valid blur transfer function is

$$\boxed{H(u,v) = \frac{1 - e^{-j2\pi vbT_1}}{j2\pi vb} + e^{-j2\pi vbT_1}\frac{1 - e^{-j2\pi uaT_2}}{j2\pi ua}}$$

where b is the first-segment velocity in the y direction and a is the second-segment velocity in the x direction.

Problem 5: Wiener Filter Expression (20 pts)

Assuming that

$$H(u,v) = 2\pi\sigma^2 e^{-2\pi^2\sigma^2(u^2+v^2)}$$

and the ratio of power spectra of the noise and undegraded signal is a constant K, give the expression for a Wiener filter.

Step 1: Wiener Filter Formula

The frequency-domain Wiener filter is

$$W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + \dfrac{S_\eta(u,v)}{S_f(u,v)}}$$

Given $\dfrac{S_\eta}{S_f} = K$ (constant),

$$W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + K}$$

Step 2: Substitute Given $H(u,v)$

Since the given $H(u,v)$ is real and positive, $H^* = H$ and

$$|H|^2 = \left(2\pi\sigma^2\right)^2 e^{-4\pi^2\sigma^2(u^2+v^2)}$$

Thus

$$W(u,v) = \frac{2\pi\sigma^2 e^{-2\pi^2\sigma^2(u^2+v^2)}}{\left(2\pi\sigma^2\right)^2 e^{-4\pi^2\sigma^2(u^2+v^2)} + K}$$

Results

$$\boxed{W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + K} = \frac{2\pi\sigma^2 e^{-2\pi^2\sigma^2(u^2+v^2)}}{4\pi^2\sigma^4 e^{-4\pi^2\sigma^2(u^2+v^2)} + K}}$$